ME315 - Lab10

```
reticulate::py_require(c("polars", "adbc-driver-sqlite"))
```

1

```
import polars as pl
import sqlite3


basics = pl.read_csv(
  "./title.basics0.tsv.gz",
  separator = "\t",
  null_values = "\\N",
  infer_schema_length = 10000
)

ratings = pl.read_csv(
  "./title.ratings.tsv.gz",
  separator = "\t",
  null_values = "\\N",
  infer_schema_length = 10000
)

principals = pl.read_csv(
  "./title.principals0.tsv.gz",
  separator = "\t",
  null_values = "\\N",
  infer_schema_length = 10000
)
```

```
import adbc_driver_manager.dbapi


conn = sqlite3.connect("./movies.sqlite3")

basics.write_database(
  table_name = "basics",
  connection = "sqlite:///movies.sqlite3",
  engine = "adbc",
  if_table_exists = "replace"
)
```

```
11144942
```

```
ratings.write_database(
  table_name = "ratings",
  connection = "sqlite:///movies.sqlite3",
  engine = "adbc",
  if_table_exists = "replace"
)
```

```
1484615
```

```
principals.write_database(
  table_name = "principals",
  connection = "sqlite:///movies.sqlite3",
  engine = "adbc",
  if_table_exists = "replace"
)
```

```
88359622
```

2

(Utilizando SQL, responda): Quais são os 5 filmes com as maiores notas (averageRating)? Apresente uma solução capaz de desempatar os filmes baseando-se no número de votos recebidos.

```
pl.read_database(r'''
  SELECT b.primaryTitle AS TITULO, r.averageRating AS NOTA, r.numVotes AS VOTANTES FROM basi
  JOIN ratings AS r ON b.tconst = r.tconst
  WHERE b.titleType = 'movie'
  ORDER BY r.averageRating DESC, r.numVotes DESC
  LIMIT 5;
  ''', conn)
```

```
shape: (5, 3)

 TITULO                      NOTA   VOTANTES
 ---                         ---    ---
 str                         f64    i64

 Kaveri                      10.0   1023
 Kurukku                     10.0   451
 Jedal Dar Omghe 30 Metri    10.0   142
 Sargashte                   10.0   134
 Gorgeous Rascal             10.0   115
```

3

(Utilizando SQL, responda): Qual é o gênero mais frequente entre os filmes com nota maior que 8?

```
pl.read_database(r'''
  SELECT g.value AS GENERO, COUNT(*) AS FREQ
  FROM basics AS b
  JOIN ratings AS r ON b.tconst = r.tconst
  JOIN json_each('["' || REPLACE(b.genres, ',', '","') || '"]') AS g
  WHERE r.averageRating > 8 AND b.titleType = 'movie'
  GROUP BY GENERO
  ORDER BY FREQ DESC
  LIMIT 1;
  ''',conn)
```

```
shape: (1, 2)

 GENERO       FREQ
```

```
  ---           ---
  str           i64

  Documentary   10564
```

4

(Utilizando SQL, responda): Quais são os 3 atores/atrizes que mais participaram de filmes com nota maior que 7.5?

```
pl.read_database(r'''
  SELECT p.nconst AS ID, COUNT(*) AS FREQ FROM principals AS p
  JOIN ratings AS r ON p.tconst = r.tconst
  JOIN basics AS b ON p.tconst = b.tconst
  WHERE r.averageRating > 7.5 AND p.category = 'actor' AND b.titleType = 'movie'
  GROUP BY ID
  ORDER BY FREQ DESC
  LIMIT 3;
  ''', conn)
```

```
shape: (3, 2)

  ID          FREQ
  ---         ---
  str         i64

  nm0004660   231
  nm0595934   155
  nm3183374   124
```

```
Sys.setenv(TZ="America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-10-16 17:00:22 -03"
```